

COEN/ELEC 490 Project Proposal

2026-2027

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Project Title: Solar-Powered Off-Grid Charging System for a 48 V Light EV for Emergency Power

Description:

This project aims to provide clean energy for off-grid charging of an electric three-wheeler, a small 48 V passenger vehicle. The vehicle is charged entirely from solar energy. This removes the extra stress vehicle charging imposes on the grid and solves the problem of routes with no charging stations. Thence the vehicle battery can be used to charge emergency loads like phones etc.

A photovoltaic array feeds a converter that charges the 48 V battery pack directly, without a grid connection. The battery pack then can supply light loads.

The students must carry out the complete design:

- ∄ **Hardware:** selection of the converter topology and of the electronic components (semiconductors, inductor, capacitors, sensors, protections);
- ∄ **Software:** selection of the microcontroller and implementation of the MPPT.
- ∄ **Design choice to justify:** the topology (buck, boost) must be justified from the measured PV module curves and from the battery voltage window.

The project ends with a prototype demonstration, where students visualize and explain the operating waveforms (PV voltage and current, switching node, inductor current, battery charging current).

Student pre-requisites:

Electrical circuits, power electronics, renewable energy, MATLAB/Simulink, and basic mathematics.

Required Resources:

All required test equipment is available in the laboratory.

- ∄ **Test equipment:** oscilloscope with differential voltage and current probes, DC power supply or PV emulator, electronic load, multimeters, soldering station.
- ∄ **Software: PSIM or MATLAB/Simulink with Simscape Electrical and Embedded Coder; Code Composer Studio.**

- € **Hardware:** PV modules, TI C2000 microcontroller board (F28379D), 48 V battery pack, gate drivers, PCB, magnetics and enclosure.
- € **Space:** Capstone Laboratory, Hall Building.

Number of Students: 6

Planning:

Event	Deadline / Time
Phase 1 (Selection and Planning)	
Project Proposal Submission	Monday, September 14, 2026 – 11:59 p.m.
Project Selection and Team Formation	Friday, September 18, 2026 – 11:59 p.m.
Phase 1 Report Submission	Sunday, October 4, 2026 – 11:59 p.m.
Phase 1 Presentation	Week of October 5, 2026
Phase 2 (Design)	
Final Design Approval	Friday, November 27, 2026 – 5:00 p.m.
Phase 2 Report Submission	Sunday, November 29, 2026 – 11:59 p.m.
Phase 2 Presentation	Week of November 30, 2026
Phase 3 (Prototype)	
In-lab Pre-Demo	Week of February 15, 2027
Phase 4 (Final Product)	
In-lab Demo	Week of March 22, 2027
Final Report Submission	Friday, April 2, 2027 – 11:59 p.m.
Final Poster Presentation	TBA

Project work in each phase:

- € **Phases 1 and 2:** design of the power converter and the control, with simulation in MATLAB/Simulink.
- € **Phase 3:** PCB design, soldering of the components, and testing of the control algorithm on a microcontroller.
- € **Phase 4:** Test of the prototype under real sunlight and final adjustments.